



C. V. Raman Global University, Bhubaneswar
Department of Mechanical Engineering
Introduction to Mechanical Engineering (IME)
Module-2

Introduction to Mechanical Engineering (ME131)

Module 2

(09 hours)

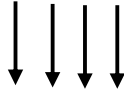
Parallel and Distributed forces: Parallel forces in a plane, Distributed Parallel forces in a plane, Couple, Resolution of a force into a force and a couple, Moment of a couple.

Centroid and Moment of Inertia: Determination of center of gravity, Center of mass, and centroid by direct integration and by the method of composite bodies, Area moment of inertia of composite plane figures and mass moment of inertia, Radius of gyration, Perpendicular and parallel axis theorem, Pappus theorems, Polar moment of inertia.



2.1 PARALLEL FORCES ON A PLANE

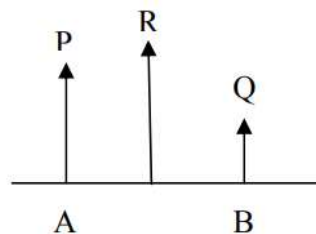
Like parallel forces: Coplanar parallel forces when act in the same direction.



Unlike parallel forces: Coplanar parallel forces when act in different direction.



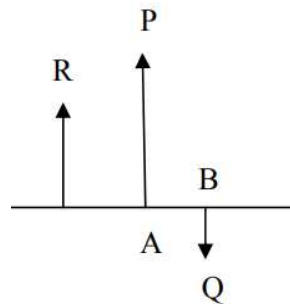
Resultant of like parallel forces:



Let P and Q are two like parallel forces act at points A and B.

$$R = P + Q$$

Resultant of unlike parallel forces:



$$R = P - Q$$

R is in the direction of the force having greater magnitude.

2.2 COUPLE

Two parallel forces equal in magnitude and opposite in direction and separated by a definite distance are said to form a couple.

- The sum of the forces forming a couple in any direction is zero, which means the translatory effect of the couple is zero.



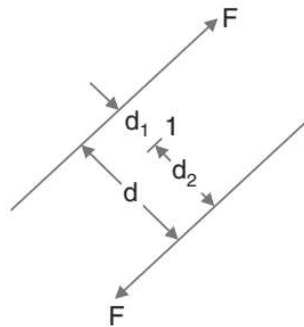
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- The rotational effect (moment) of a couple about any point is a constant and it is equal to the product of the magnitude of the forces and the perpendicular distance between the two forces.

Consider the following cases

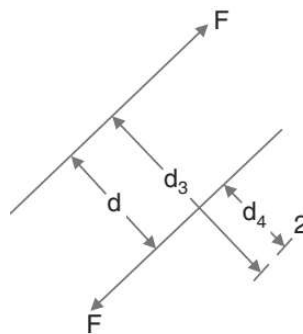
1.



Let the moment be M_1 . Then,

$$M_1 = Fd_1 + Fd_2 = F(d_1 + d_2)$$
$$M_1 = Fd$$

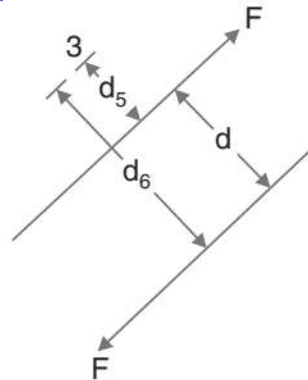
2.



Let M_2 be the moment. Then,

$$M_2 = Fd_3 - Fd_4 = F(d_3 - d_4)$$
$$M_2 = Fd$$

3.



Similarly,

$$M_3 = Fd_6 - Fd_5 = F(d_6 - d_5)$$
$$M_3 = Fd.$$

It can be observed that moment of a couple about any point is the same. Now we can summarise the following properties of a couple:

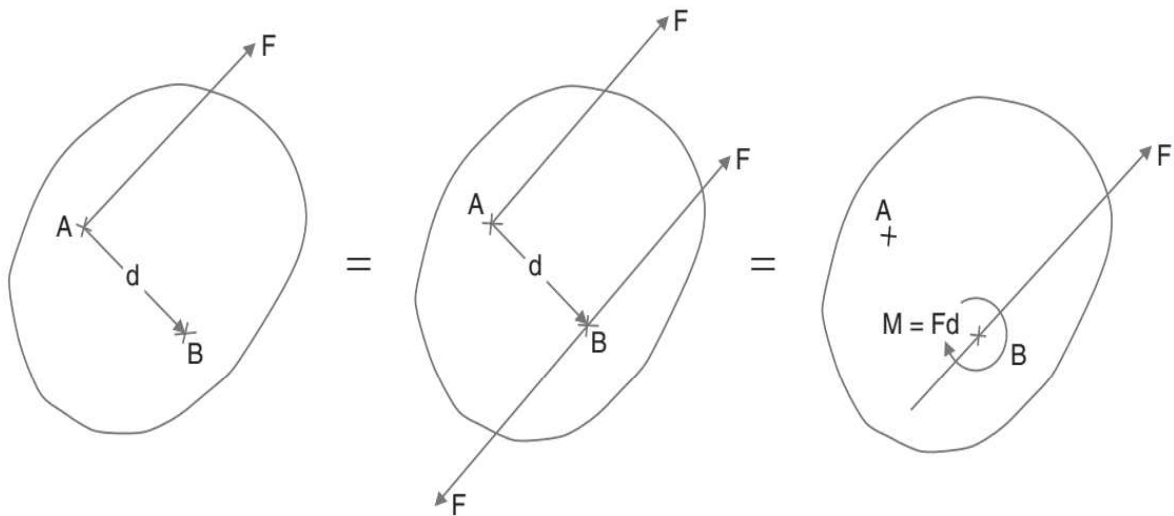
- A couple consists of a pair of equal and opposite parallel forces which are separated by a definite distance.
- The translatory effect of a couple on the body is zero.

Since only effect of a couple is the moment, which is the same at all points, the effect of a couple is unchanged if,

- The couple is rotated through any angle.
- The couple is shifted to any other position.
- The couple is replaced by another pair of forces whose rotational effect is the same.

Resolution of a force into a force and a couple

In Fig. (a), F is a force acting on a body at point A. By applying equal and opposite forces of magnitude F at B, the effect of system of forces is unaltered (Law of superposition). Hence the system of forces in Fig. (b) is the same as the system given in Fig. (a). Now the original force F at A and the opposite force F at B form a couple moment of magnitude $F \times d$. Hence the system in Fig. (b) is equal to the system of forces in Fig. (c). Thus the given force F at A is equal to a force F at B and a moment Fd .



2.3 CENTRE OF GRAVITY

A body is composed of an infinite number of particles of differential size, and so if the body is located within a gravitational field, then each of these particles will have a weight dW . These weights will form a parallel force system, and the resultant of this system is the total weight of the body, which passes through a single point called the *center of gravity* (G).

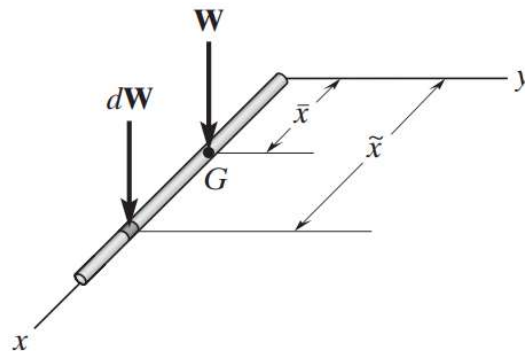


Figure 1

To show how to determine the location of the center of gravity, consider the rod in Fig. 1, where the segment having the weight dW is located at the arbitrary position $\tilde{x}$. The total weight of the rod is the sum of the weights of all of its particles, that is

$$W = \int dW$$



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The location of the center of gravity, measured from the y axis, is determined by equating the moment of W about the y axis, to the sum of the moments of the weights of all its particles about this same axis. Therefore,

$$(M_R)_y = \Sigma M_y;$$

$$\bar{x}W = \int \tilde{x}dW$$

$$\bar{x} = \frac{\int \tilde{x} dW}{\int dW}$$

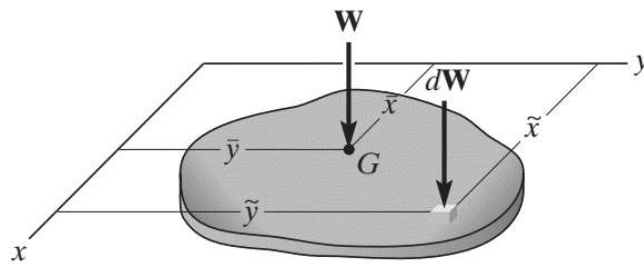


Figure 2

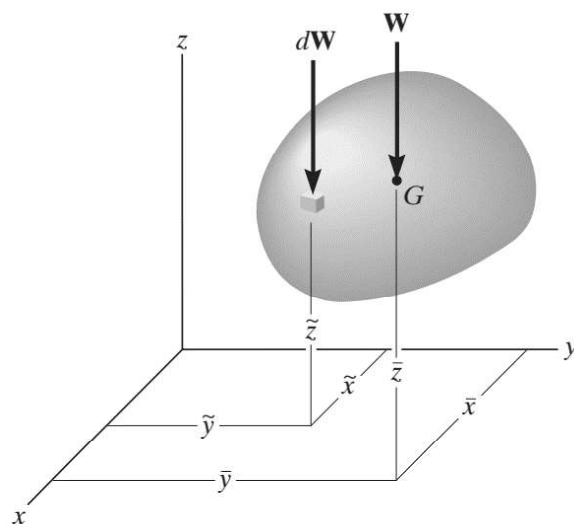


Figure 3



In a similar manner, if the body represents a plate, Fig. 2, then a moment balance about the x and y axes would be required to determine the location $(\bar{x}, \bar{y})$ of point G . Finally, we can generalize this idea to a three-dimensional body, Fig. 3, and perform a moment balance about all three axes to locate G for any rotated position of the axes. This results in the following equations.

$$\bar{x} = \frac{\int \tilde{x} dW}{\int dW} \quad \bar{y} = \frac{\int \tilde{y} dW}{\int dW} \quad \bar{z} = \frac{\int \tilde{z} dW}{\int dW} \quad (1)$$

where $\bar{x}, \bar{y}, \bar{z}$ are the coordinates of the center of gravity G

$\tilde{x}, \tilde{y}, \tilde{z}$ are the coordinates of an arbitrary particle in the body

2.4 CENTER OF MASS OF A BODY

In order to study the dynamic response or accelerated motion of a body, it becomes important to locate the body's *center of mass* C_m , Fig. 4. This location can be determined by substituting $dW = g dm$ into Eq.1. Provided g is constant, it cancels out, and so

$$\bar{x} = \frac{\int \tilde{x} dm}{\int dm} \quad \bar{y} = \frac{\int \tilde{y} dm}{\int dm} \quad \bar{z} = \frac{\int \tilde{z} dm}{\int dm} \quad (2)$$

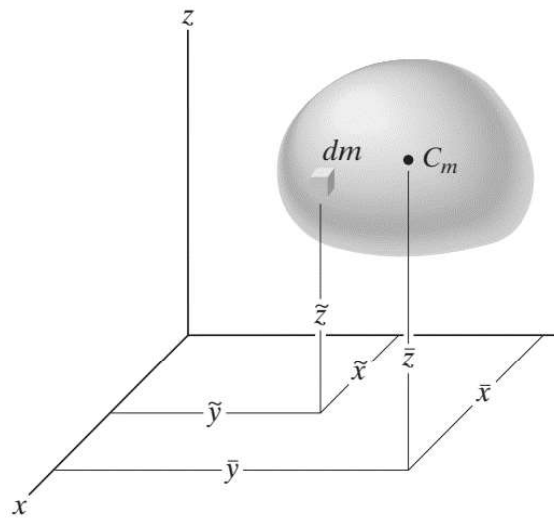


Figure 4



2.5 CENTROID OF A VOLUME

If the body in Fig. 5 is made from a homogeneous material, then its density ρ will be constant. Therefore, a differential element of volume dV has a mass $dm = \rho dV$. Substituting this into Eq. 2 and canceling out ρ , we obtain formulas that locate the *centroid* C or geometric center of the body, namely,

$$\bar{x} = \frac{\int_V \tilde{x} dV}{\int_V dV} \quad \bar{y} = \frac{\int_V \tilde{y} dV}{\int_V dV} \quad \bar{z} = \frac{\int_V \tilde{z} dV}{\int_V dV} \quad (3)$$

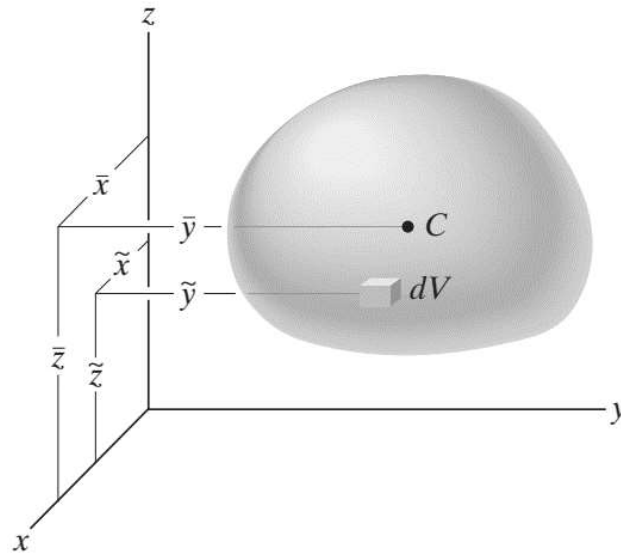


Figure 5

2.6 CENTROID OF AN AREA

If an area lies in the x - y plane and is bounded by the curve $y = f(x)$, as shown in Fig. 6, then its centroid will be in this plane and can be determined from integrals similar to Eq. 3, namely,

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA} \quad \bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA} \quad (4)$$

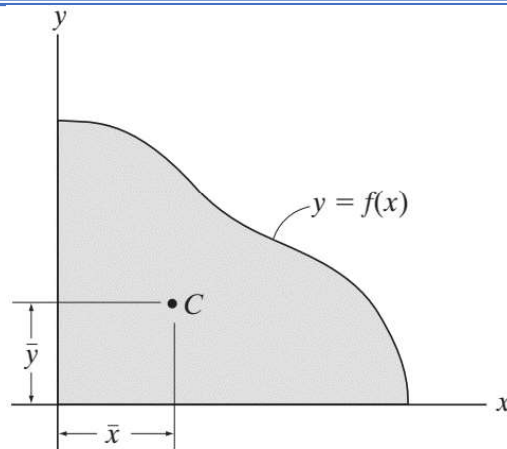


Figure 6

2.7 CENTROID OF A LINE

If a line segment (or a rod) lies within the x-y plane and it can be described by a thin curve $y = f(x)$, Fig. 7, then its centroid is determined from

$$\bar{x} = \frac{\int_L \tilde{x} dL}{\int_L dL} \quad \bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL}$$

(5)

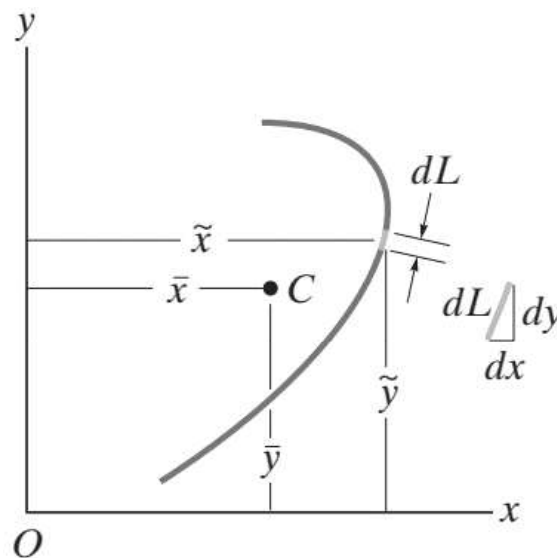


Figure 7



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Here, the length of the differential element is given by the *Pythagorean theorem*, $dL = \sqrt{(dx)^2 + (dy)^2}$, which can also be written in the form

$$\begin{aligned} dL &= \sqrt{\left(\frac{dx}{dx}\right)^2 dx^2 + \left(\frac{dy}{dx}\right)^2 dx^2} \\ &= \left(\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\right) dx \end{aligned}$$

or

$$\begin{aligned} dL &= \sqrt{\left(\frac{dx}{dy}\right)^2 dy^2 + \left(\frac{dy}{dy}\right)^2 dy^2} \\ &= \left(\sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\right) dy \end{aligned}$$

Either one of these expressions can be used; however, for application, the one that will result in a simpler integration should be selected.

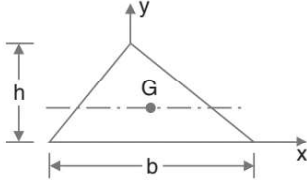
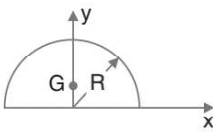
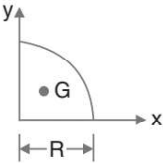
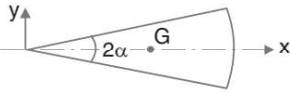
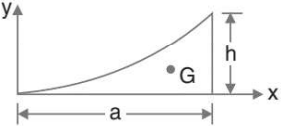
Important Points

- The centroid represents the geometric center of a body. This point coincides with the center of mass or the center of gravity only if the material composing the body is uniform or homogeneous.
- Formulas used to locate the center of gravity or the centroid simply represent a balance between the sum of moments of all the parts of the system and the moment of the “resultant” for the system.
- In some cases, the centroid is located at a point that is not on the object, as in the case of a ring, where the centroid is at its center. Also, this point will lie on any axis of symmetry for the body



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Centroid of some Common Figures

Shape	Figure	$\bar{x}$	$\bar{y}$	Area
Triangle		—	$\frac{h}{3}$	$\frac{bh}{2}$
Semicircle		0	$\frac{4R}{3\pi}$	$\frac{\pi R^2}{2}$
Quarter circle		$\frac{4R}{3\pi}$	$\frac{4R}{3\pi}$	$\frac{\pi R^2}{4}$
Sector of a circle		$\frac{2R}{3\alpha} \sin \alpha$	0	αR^2
Parabolic Spandrel		$\frac{3a}{4}$	$\frac{3h}{10}$	$\frac{ah}{3}$

2.8 COMPOSITE BODIES

A composite body consists of a series of connected “simpler” shaped bodies, which may be rectangular, triangular, semicircular, etc. Such a body can often be sectioned or divided into its composite parts and, provided the weight and location of the center of gravity of each of these parts are known, we can then eliminate the need for integration to determine the center of gravity for the entire body. The method for doing this follows the same procedure outlined earlier; however, rather than account for an infinite number of differential weights, we have instead a finite number of weights. Therefore,



$$\bar{x} = \frac{\sum \tilde{x}W}{\sum W} \quad \bar{y} = \frac{\sum \tilde{y}W}{\sum W} \quad \bar{z} = \frac{\sum \tilde{z}W}{\sum W}$$

(6)

where $\bar{x}, \bar{y}, \bar{z}$ represent the coordinates of the center of gravity of the composite body

$\tilde{x}, \tilde{y}, \tilde{z}$ represent the coordinates of the center of gravity of each part of the body

$\sum W$ is the sum of the weights of all the composite parts of the body

When the body has a constant density or specific weight, the center of gravity coincides with the centroid of the body. The centroid for composite lines, areas, and volumes can be found using relations analogous to Eq. 6; however, the W 's are replaced by L 's, A 's, and V 's, respectively.

2.9 MOMENT OF INERTIA

Consider the plate in Fig. 8, which is submerged in a fluid and subjected to the pressure p . According to Pascal's law, this pressure varies linearly with depth, such that $p = \gamma y$, where γ is the specific weight of the fluid. Thus, the force acting on the differential area dA of the plate is $dF = p dA = (\gamma y)dA$.

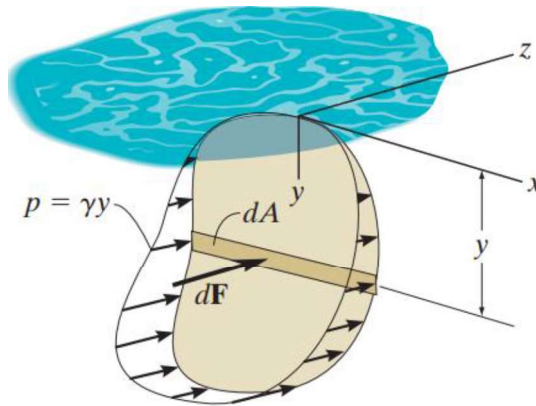


Figure 8

The moment of this force about the x axis is therefore $dM = y dF = \gamma y^2 dA$, and so integrating dM over the entire area of the plate yields $M = \gamma \int y^2 dA$. The integral $y^2 dA$ is sometimes referred to as the “second moment” of the area about an axis (the x axis), but more often it is called the *moment of inertia of the area*.

By definition, the moments of inertia of a differential area dA about the x and y axes are $dI_x = y^2 dA$ and $dI_y = x^2 dA$, respectively, Fig. 9. For the entire area A the moments of inertia are



determined by integration; i.e.,

$$\begin{aligned} I_x &= \int_A y^2 dA \\ I_y &= \int_A x^2 dA \end{aligned} \quad (7)$$

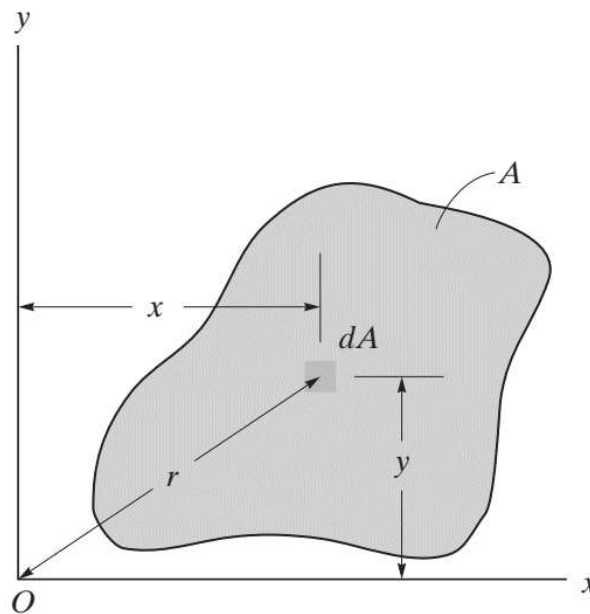


Figure 9

We can also formulate this quantity for dA about the “pole” O or z axis, Fig. 9. This is referred to as the polar moment of inertia. It is defined as $dJ_O = r^2 dA$, where r is the perpendicular distance from the pole (z axis) to the element dA . For the entire area the polar moment of inertia is

$$J_O = \int_A r^2 dA = I_x + I_y \quad (8)$$

2.10 PARALLEL-AXIS THEOREM

The parallel-axis theorem can be used to find the moment of inertia of an area about any axis that is parallel to an axis passing through the centroid and about which the moment of inertia is known.

Let's choose a differential element dA located at an arbitrary distance y' from the centroidal x' axis. If the distance between the parallel x and x' axis is d_y , then the moment of



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inertia of dA about the x axis is $dI_x = (y' + d_y)^2 dA$. For the entire area,

$$\begin{aligned} I_x &= \int_A (y' + d_y)^2 dA \\ &= \int_A y'^2 dA + 2d_y \int_A y' dA + d_y^2 \int_A dA \end{aligned}$$

The first integral represents the moment of inertia of the area about the centroidal axis, $\bar{I}_{x'}$. The second integral is zero since the x' axis passes through the area's centroid C ; i.e., $\int y' dA = \bar{y}' \int dA = 0$ since $\bar{y}' = 0$. Since the third integral represents the total area A , the final result is therefore

$$I_x = \bar{I}_{x'} + Ad_y^2 \quad (9)$$

A similar expression can be written for I_y ; i.e.,

$$I_y = \bar{I}_{y'} + Ad_x^2 \quad (10)$$

And finally, for the polar moment of inertia, since $\bar{J}_C = \bar{I}_{x'} + \bar{I}_{y'}$ and $d^2 = d_x^2 + d_y^2$, we have

$$J_O = \bar{J}_C + Ad^2 \quad (11)$$

The form of each of these three equations states that the moment of inertia for an area about an axis is equal to its moment of inertia about a parallel axis passing through the area's centroid plus the product of the area and the square of the perpendicular distance between the axes.

2.11 RADIUS OF GYRATION

Radius of gyration of a body about the axis of rotation is defined as the radial distance to a point which would have a moment of inertia the same as the body's actual distribution of mass, if the total mass of the body were concentrated there. The radius of gyration of an area about an axis has units of length and is a quantity that is often used for the design of columns in structural mechanics.

$$k_x = \sqrt{\frac{I_x}{A}} \quad k_y = \sqrt{\frac{I_y}{A}} \quad k_O = \sqrt{\frac{J_O}{A}} \quad (12)$$



Important Points

- The moment of inertia is a geometric property of an area that is used to determine the strength of a structural member or the location of a resultant pressure force acting on a plate submerged in a fluid. It is sometimes referred to as the second moment of the area about an axis, because the distance from the axis to each area element is squared.
- If the moment of inertia of an area is known about its centroidal axis, then the moment of inertia about a corresponding parallel axis can be determined using the parallel-axis theorem.

2.12 MOMENTS OF INERTIA FOR COMPOSITE AREAS

A composite area consists of a series of connected “simpler” parts or shapes, such as rectangles, triangles, and circles. Provided the moment of inertia of each of these parts is known or can be determined about a common axis, then the moment of inertia for the composite area about this axis equals the algebraic sum of the moments of inertia of all its parts.

Procedure for Analysis

The moment of inertia for a composite area about a reference axis can be determined using the following procedure.

Composite Parts

- Using a sketch, divide the area into its composite parts and indicate the perpendicular distance from the centroid of each part to the reference axis.

Parallel-Axis Theorem

- If the centroidal axis for each part does not coincide with the reference axis, the parallel-axis theorem, $I = \bar{I} + Ad^2$, should be used to determine the moment of inertia of the part about the reference axis.

Summation

- The moment of inertia of the entire area about the reference axis is determined by summing the results of its composite parts about this axis.



- If a composite part has an empty region (hole), its moment of inertia is found by subtracting the moment of inertia of this region from the moment of inertia of the entire part including the region.

2.13 MASS MOMENT OF INERTIA

The mass moment of inertia of a body is a measure of the body's resistance to angular acceleration. It is used in dynamics to study rotational motion. Consider the rigid body shown in Fig. 10. The mass moment of inertia of the body about the z axis is

$$I = \int_m r^2 dm \quad (13)$$

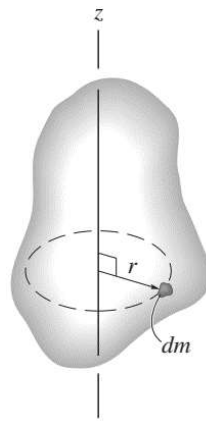


Fig. 10

Here r is the perpendicular distance from the axis to the arbitrary element dm . Since the formulation involves r , the value of I is unique for each axis about which it is computed. The axis which is generally chosen, however, passes through the body's mass center G . If the body consists of material having a density ρ , then $dm = \rho dV$. Substituting this into Eq. 13, the body's moment of inertia is then computed using volume elements for integration; i.e.,

$$I = \int_V r^2 \rho dV \quad (14)$$

For most applications, ρ will be a constant, and so this term may be factored out of the integral, and the integration is then purely a function of geometry.



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$$I = \rho \int_V r^2 dV \quad (15)$$

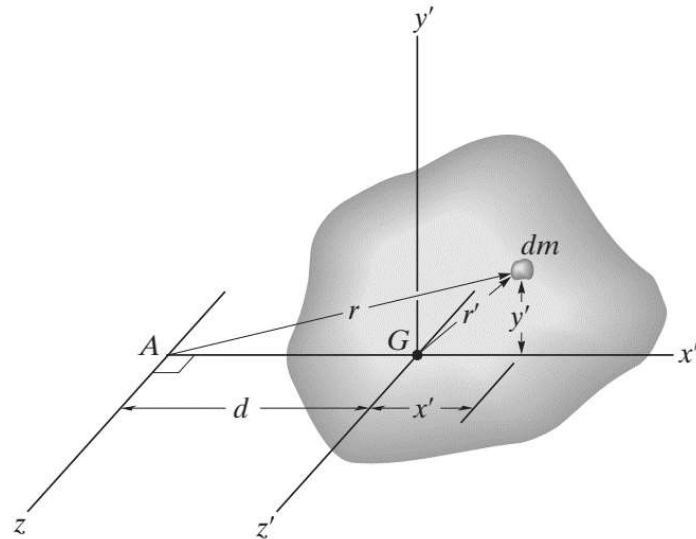


Figure 11



Parallel-Axis Theorem. If the moment of inertia of the body about an axis passing through the body's mass center is known, then the moment of inertia about any other *parallel axis* can be determined by using the *parallel-axis theorem*. To derive this theorem, consider the body shown in Fig. 11. The z' axis passes through the mass center G , whereas the corresponding *parallel* z axis lies at a constant distance d away. Selecting the differential element of mass dm , which is located at point (x', y') , and using the Pythagorean theorem, $r^2 = (d + x')^2 + y'^2$, the moment of inertia of the body about the z axis is

$$\begin{aligned} I &= \int_m r^2 dm = \int_m [(d + x')^2 + y'^2] dm \\ &= \int_m (x'^2 + y'^2) dm + 2d \int_m x' dm + d^2 \int_m dm \end{aligned}$$

Since $r'^2 = x'^2 + y'^2$, the first integral represents I_G . The second integral is equal to *zero*, since the z' axis passes through the body's mass center, i.e., $\int x' dm = \bar{x} \int dm = 0$ since $\bar{x} = 0$. Finally, the third integral is the total mass m of the body. Hence, the moment of inertia about the z axis becomes

$$I = I_G + md^2 \quad (16)$$

where

I_G = moment of inertia about the z' axis passing through the mass center G

m = mass of the body

d = distance between the parallel axes



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Radius of Gyration. Occasionally, the moment of inertia of a body about a specified axis is reported in handbooks using the *radius of gyration*, k . This value has units of length, and when it and the body's mass m are known, the moment of inertia can be determined from the equation

$$I = mk^2 \quad \text{or} \quad k = \sqrt{\frac{I}{m}} \quad (16)$$

Note the *similarity* between the definition of k in this formula and r in the equation $dI = r^2 dm$, which defines the moment of inertia of a differential element of mass dm of the body about an axis.

2.14 COMPOSITE BODIES

If a body is constructed from a number of simple shapes such as disks, spheres, and rods, the moment of inertia of the body about any axis z can be determined by adding algebraically the moments of inertia of all the composite shapes calculated about the same axis. Algebraic addition is necessary since a composite part must be considered as a negative quantity if it has already been included within another part—as in the case of a “hole” subtracted from a solid plate. Also, the parallel-axis theorem is needed for the calculations if the center of mass of each composite part does not lie on the z axis.

2.15 THEOREMS OF PAPPUS AND GULDINUS

The two theorems of Pappus and Guldinus are used to find the *surface area* and *volume* of any body of revolution. They were first developed by Pappus of Alexandria during the fourth century a.d. and then restated at a later time by the Swiss mathematician Paul Guldin or Guldinus (1577–1643).

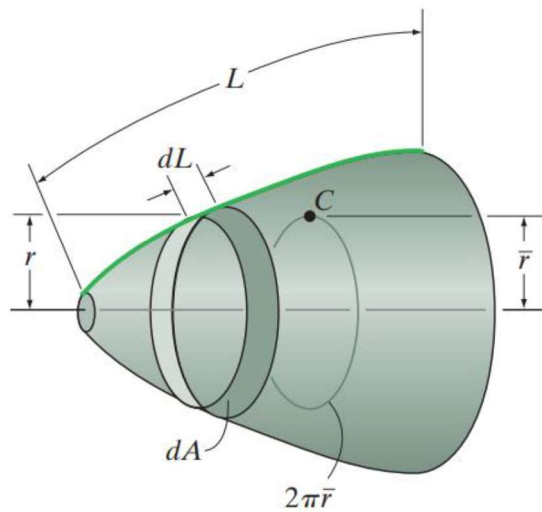


Figure 12

2.14.1 Surface Area

If we revolve a plane curve about an axis that does not intersect the curve, we will generate a surface area of revolution. For example, the surface area in Fig. 12 is formed by revolving the curve of length L about the horizontal axis. To determine this surface area, we will first consider the differential line element of length dL . If this element is revolved 2π radians about the axis, a ring having a surface area of $dA = 2\pi r dL$ will be generated. Thus, the surface area of the entire body is $A = 2\pi \int r dL$. Since $\int r dL = \bar{r}L$, then $A = 2\pi \bar{r}L$. If the curve is revolved only through an angle θ (radians), then

$$A = \theta \bar{r}L \quad (17)$$

where

A = surface area of revolution

θ = angle of revolution measured in radians, $\theta \leq 2\pi$

$\bar{r}$ = perpendicular distance from the axis of revolution to the centroid of the generating curve

L = length of the generating curve

Therefore, the first theorem of Pappus and Guldinus states that the area of a surface of revolution equals the product of the length of the generating curve and the distance travelled by the centroid of the curve in generating the surface area.



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2.14.2 Volume

A volume can be generated by revolving a plane area about an axis that does not intersect the area. For example, if we revolve the shaded area A in Fig. 13 about the horizontal axis, it generates the volume shown. This volume can be determined by first revolving the differential element of area dA 2π radians about the axis, so that a ring having the volume $dV = 2\pi r dA$ is generated. The entire volume is then $V = 2\pi \int r dA$. However, $\int r dA = \bar{r}A$, so that $V = 2\pi \bar{r}A$. If the area is only revolved through an angle θ (radians), then

$$V = \theta \bar{r}A \quad (18)$$

where

V = volume of revolution

θ = angle of revolution measured in radians, $\theta \leq 2\pi$

$\bar{r}$ = perpendicular distance from the axis of revolution to the centroid of the generating area

A = generating area

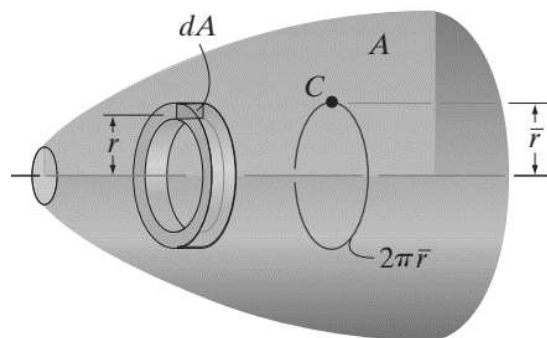


Figure 13

Therefore, the second theorem of Pappus and Guldinus states that the volume of a body of revolution equals the product of the generating area and the distance travelled by the centroid of the area in generating the volume.

2.14.3 Composite Shapes

We may also apply the above two theorems to lines or areas that are composed of a series of composite parts. In this case the total surface area or volume generated is the addition of the surface areas or volumes generated by each of the composite parts. If the perpendicular distance from the axis of revolution to the centroid of each composite part is $\bar{r}$, then



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$$A = \theta \sum(\tilde{r}L) \quad (19)$$

and

$$V = \theta \sum(\tilde{r}A) \quad (20)$$

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